Technical Report: AI-Powered Personal CRM Assistant

Challenge Tackled We built an AI-powered Personal CRM assistant that extracts contact details from unstructured sources like business cards, images, and PDFs, and enables users to send personalized LinkedIn connection requests. The target user is a networking professional or founder who needs to quickly digitize and manage contacts after events.

Tools / ML Models Used

- **Tesseract OCR** – for extracting text from images
- **spaCy (en_core_web_sm)** – for named entity recognition (e.g., name, email, company)
- **Flask** – backend API
- **Selenium + ChromeDriver** – to automate LinkedIn connect and messaging
- **SQLite** – lightweight local contact storage
- **Docker** – containerized environment for deployment
- **HTML/JS** – simple frontend for upload and messaging

What Worked Well

- The OCR-NLP pipeline extracted names, emails, phone numbers, and companies from images.
- The SQLite database and search endpoint enabled easy, fast lookups.

What Was Challenging

- ChromeDriver automation inside Docker was complex due to GUI dependencies; ultimately solved by testing outside container.
- CORS and network access issues between frontend and Flask backend required absolute URLs and CORS headers.
- The LinkedIn integration has issues, and would likely be resolved in a few more hours.

How We Spent Our Time

- **0–1h**: Problem scoping and API design
- **1-3h**: OCR + NLP pipeline development
- **3–5h**: Flask API + SQLite integration
- **5–9h**: Debugging LinkedIn Selenium automation
- **9–11h**: Dockerization + debugging
- **11–12h**: Frontend UI + polish + testing + documentation
- If we had 24 more hours, we'd resolve the Linkedin integration, and add voice-to-contact card support.
